RA #3

Wednesday, September 16, 2026 12:17 PM

Colby Magee
Reading Assignment #3

CHE 4990/CHE5990 **Fall 2026**

Reading Assignment 3 Questions (40 points)

Due 11.59 PM on September 16, 2026

Reading article:

The reading article for this assignment is

https://www.thechemicalengineer.com/features/the-process-scaleup-journey-part-3-tea-lca-finding-the-balance/

Question 1 – based on the class lecture (10 points)

To meet the energy needs of a data center in Stillwater, OK, the city of Stillwater hypothetically chooses to implement a new electrolyzer technology that is 20% more energy-efficient than the current power plant. Answer the questions below using the information provided above.

a. Classify each of the items as CAPEX (capital expenditure), variable OPEX (variable operating expenditure), or fixed OPEX (fixed operating expenditure) and provide a short reason.

Item	Classification	Reason
Electrolyzer purchase		
Electricity consumption by the electrolyzer		
Installation and piping		
Operator salaries		
Water consumption		
Annual insurance		
Routine maintenance		
Construction		
Feedstock consumption		
Parts inside the electrolyzer that need periodic replacement		

b. (5 points) The estimated cost of hydrogen production from this electrolyzer is $5 per kg. The table below identifies four parameters for a sensitivity analysis. Indicate the following for each parameter:
    a. Does the hydrogen cost go up or down as the parameter changes?
    b. Provide the reason(s) for this trend.

Parameter	Parameter change	Expected outcome	Reason
Electricity Price	Increases		
Electrolyzer efficiency	Increases		
Capital cost	Increases		
Capacity factor	Increases		

_____ (fill in the blank) parameter has the highest impact because _____ (fill in the blank).

Question 2 – based on the class lecture (18 points)

Imagine a hypothetical change suggested by an engineer to an energy conversion process. The table below highlights the differences between the existing and the proposed modified processes.

Metric	Existing Process	Proposed Modified Process
Energy efficiency	65%	78%
Exergy efficiency	42%	61%
Capital cost	$ 10 million	$ 16 million
Operating cost	$ 4 million per year	$ 3.2 million per year
Greenhouse gas (GHG) emissions	100 units	72 units

a. (3 pts) From a thermodynamic/exergy perspective, which process is preferable? Why?
b. (5 pts) From a TEA perspective, is the available data enough to determine if the proposed modified process is economically preferred? If not, specify which additional information is needed.
c. (5 pts) From an LCA perspective, is the available data enough to determine if the proposed modified process is environmentally better? If not, specify which additional information is needed.
d. (5 pts) If you were the engineer responsible for finally accepting the existing process or the proposed modified process, what additional information would you consider before making the final recommendation?

Question 3 – based on the reading article (12 points)

a. How did the TEA and LCA influence the FluoRok technology? Explain in your own words (6 points).
b. Direct air capture is an example where TEA and LCA may lead to conflicting conclusions. Explain in detail why LCA and TEA contradict each other (6 points).

Question 1

Item	Classification	Reason
Electrolyzer purchase	CAPEX	Single upfront cost
Electricity consumption by the electrolyzer	VAIRABLE OPEX	Electricity consumption is depended n run time and load
Installation and piping	CAPEX	Single upfront cost
Operator salaries	FIXED OPEX	Assuming pay raise is periodic and sufficiently long
Water consumption	VARIABLE OPEX	Depended on run time and load
Annual insurance	FIXED OPEX	Same insurance policy
Routine maintenance	FIXED OPEX	Implies same fix at constant intervals
Construction	CAPEX	Single upfront cost
Feedstock consumption	VARIABLE OPEX	Dependent on load and run time
arts inside the electrolyzer that need periodic replacement	VARIABLE OPEX	Periodic implies random times and random parts

Parameter	**Parameter change**	**Expected outcome for H_2 cost**	**Reason**
Electricity price	Increases	Increases	higher electricity price means higher operating cost = higher cost per kg of H_2.
Electrolyzer efficiency	Increases	Decreases	Higher efficiency = less electricity for each kg of H_2.
Capital cost	Increases	Increases	Higher capital investment means more money needed to make back from sales
Capacity factor	Increases	Decreases	The electrolyzer produces more H_2, spreading fixed costs over more kg of H_2.

Question 2

A) Modified process is better thermodynamically, because it has a higher 1st and 2nd law efficiency

B) No, it is not enough information. It list capital and operating cost but does not included information on cash flow, ROI or NPV. These factor are listed as assessment metrics in CPIs table
C) No, it is not enough information. Reading from CPI's table 1. Assessment metrics are human toxicity, ozone depletion, land use, water ETC. The proposed modification only references greenhouse gas emissions
D) I Would need to consider all factors mentioned in the previous two answers as well as the technical readiness level of the modifications

Question 3

A) The insights CPI provided to Flourok were in the areas of energy demand, hot spot and tradeoffs. The insight provided although not mentioned in detail allowed for them to increase efficiency of their fluorochemical manufacturing and advanced the systems TRL rating

B) Direct Air capture has the potential to be a negative emission system assuming power source is environmentally friendly and so has a good LCA rating. However its TEA rating is low because it requires large amount of electricity. Thus Is has a large economic barrier to entry